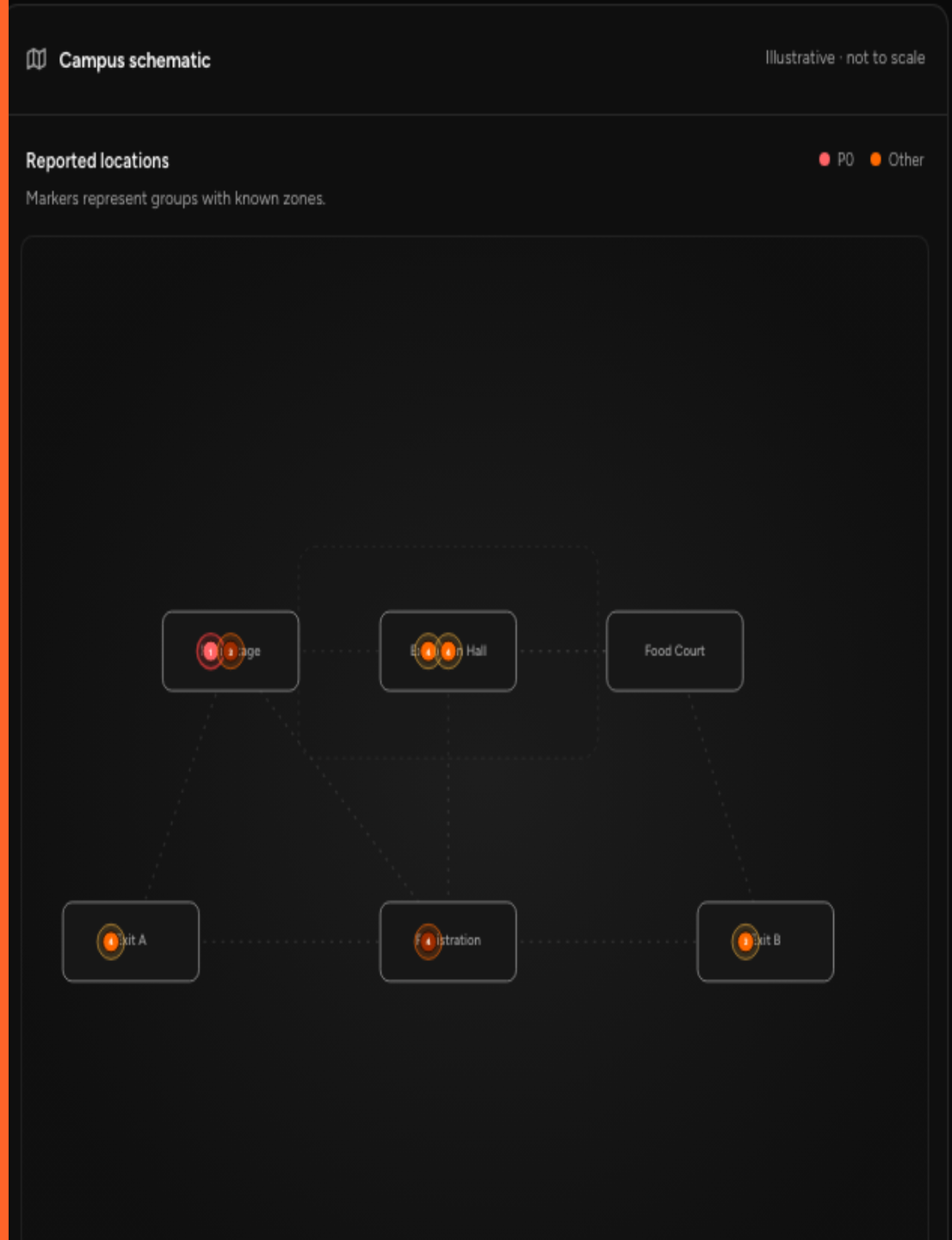
AI Innovate-A-Thon, Topic 10

Aegis

Emergency First-Response Coordination

A campus event exercise prototype

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Exit A

Three accounts of the same doorway

“The inner door at Exit A will not open.”

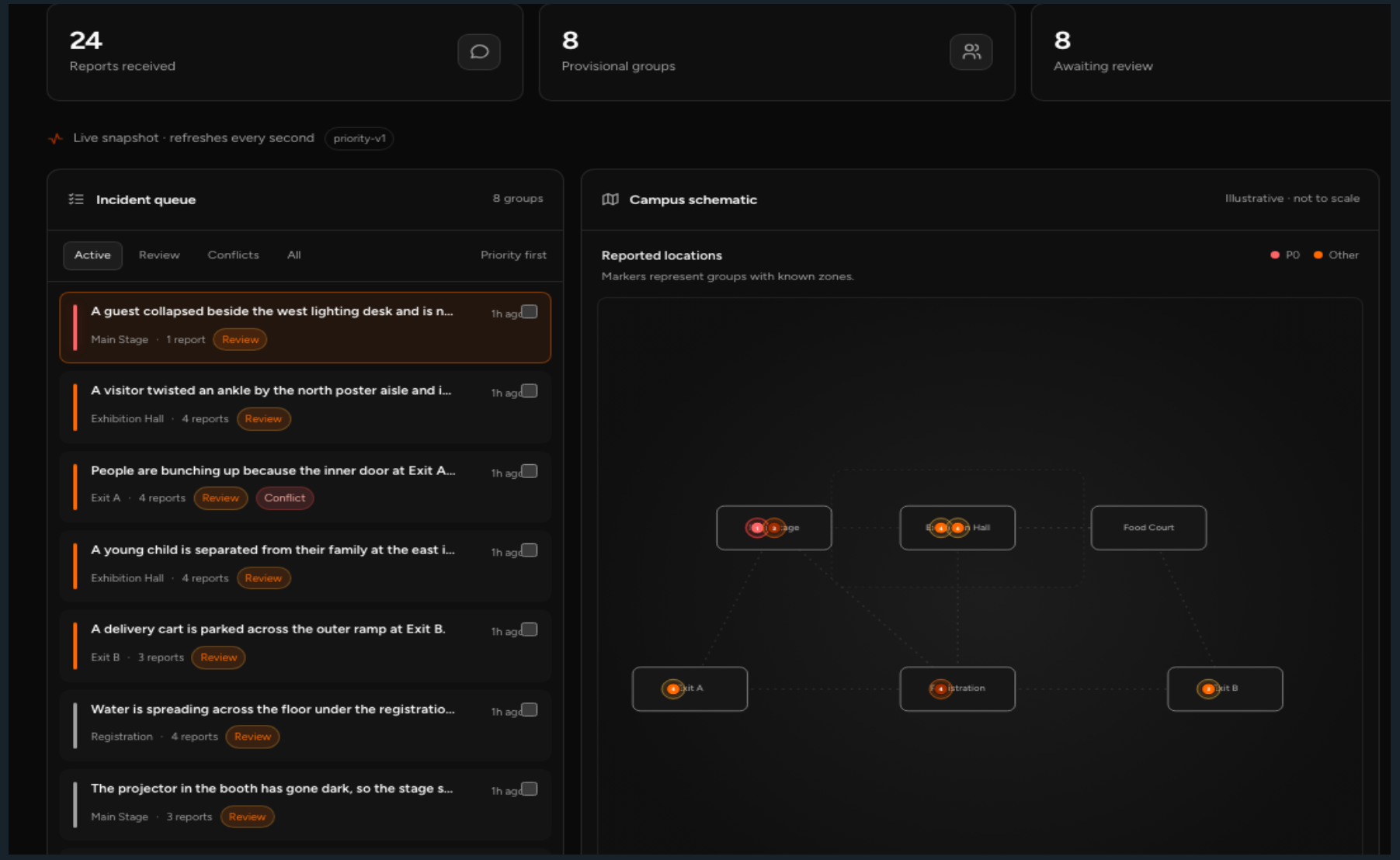
“The inside Exit A door seems jammed.”

“Exit A doorway is clear”



reports to reconcile

The incident board



24

reports received

8

provisional groups

3

conflicting pairs

The processing path

1 **Receive**

Form or replay

FastAPI + Pydantic

2 **Preserve**

Original text and receipt

SQLite transaction

3 **Analyse**

Matching evidence and priority

Python engine

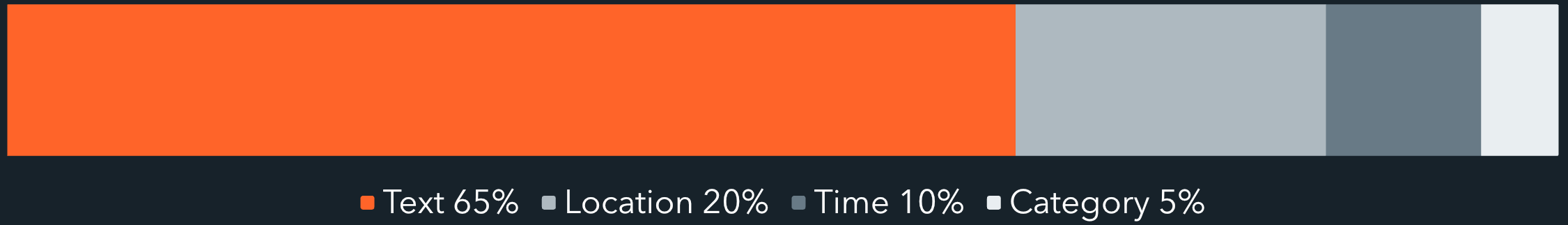
4 **Review**

Confirm or correct with a reason

React + TypeScript

Matching

Text and event context contribute to the ranking score



0.33
candidate score

0.05
anchor similarity

0.08
candidate margin

Lexical fallback gates. Compatible time and specific shared context are also required.

Priority has an explicit rule

P0 Affirmative immediate-danger phrase

P1 Blocked exit, medical concern or vulnerable missing person

P2 Facility issue with a supplied zone

? Insufficient supported information

Unassessed

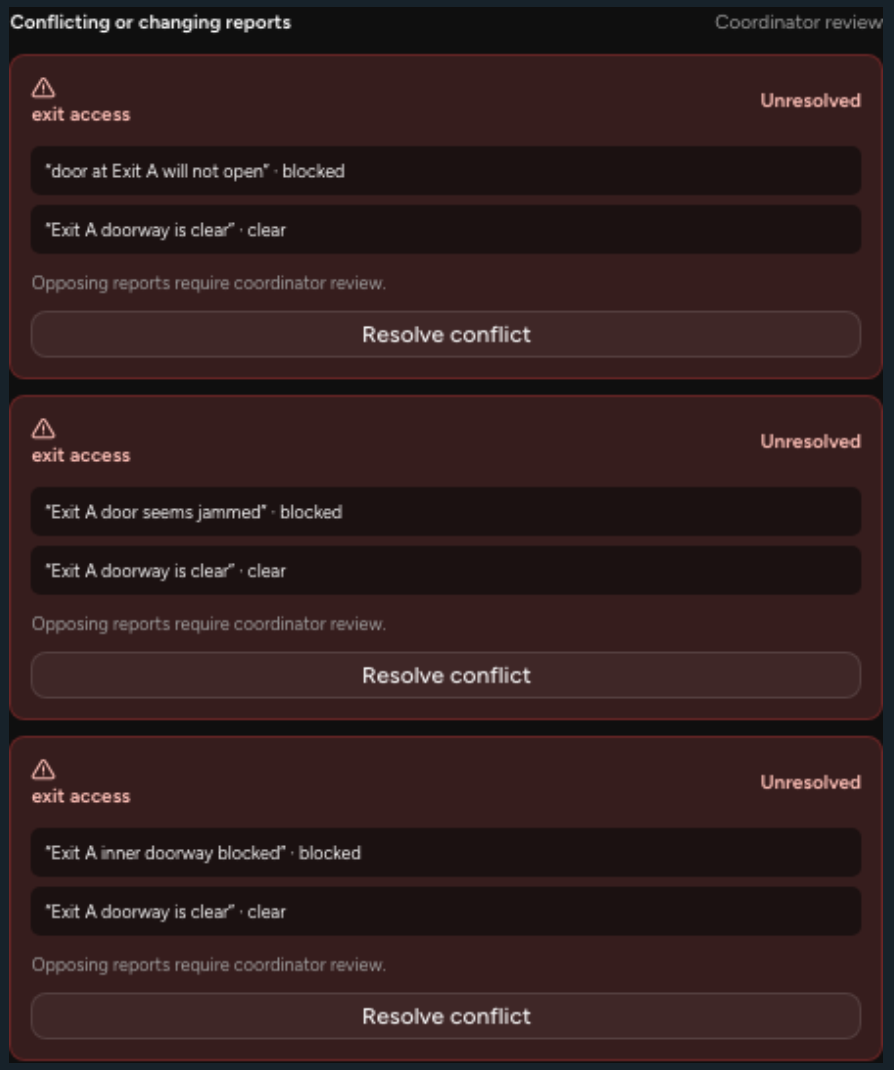
The source phrase stays visible beside the priority suggestion.

Conflicting accounts

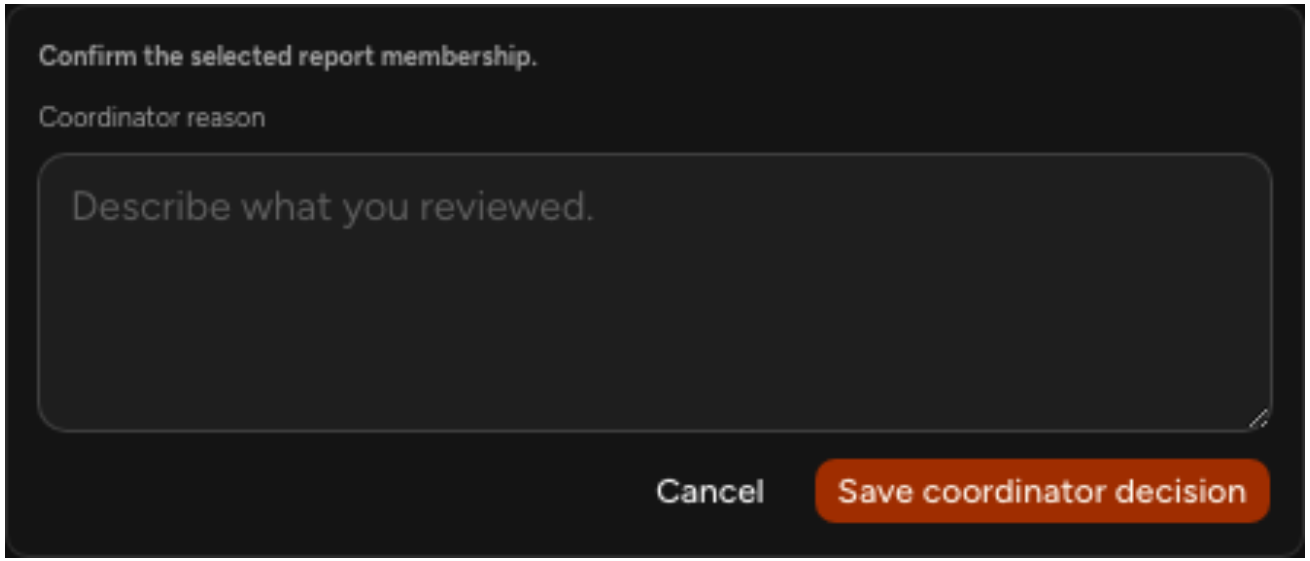
"Blocked"

"Clear"

Both accounts remain available.
The coordinator reviews the change and records a decision.



The coordinator decides



Confirm or reject
Move, merge or split

A reason is required.
Stale versions return HTTP 409.

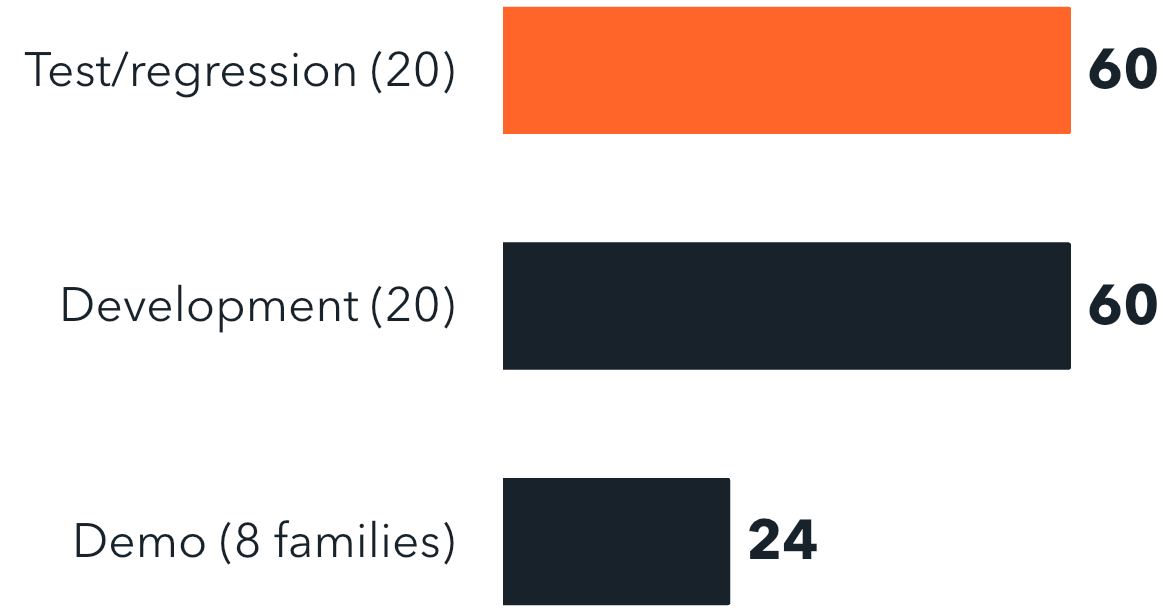
“Reviewed the Exit A accounts together. Keep the changing door status visible for follow-up.”

144

synthetic reports

Labels are stored separately from runtime inputs.

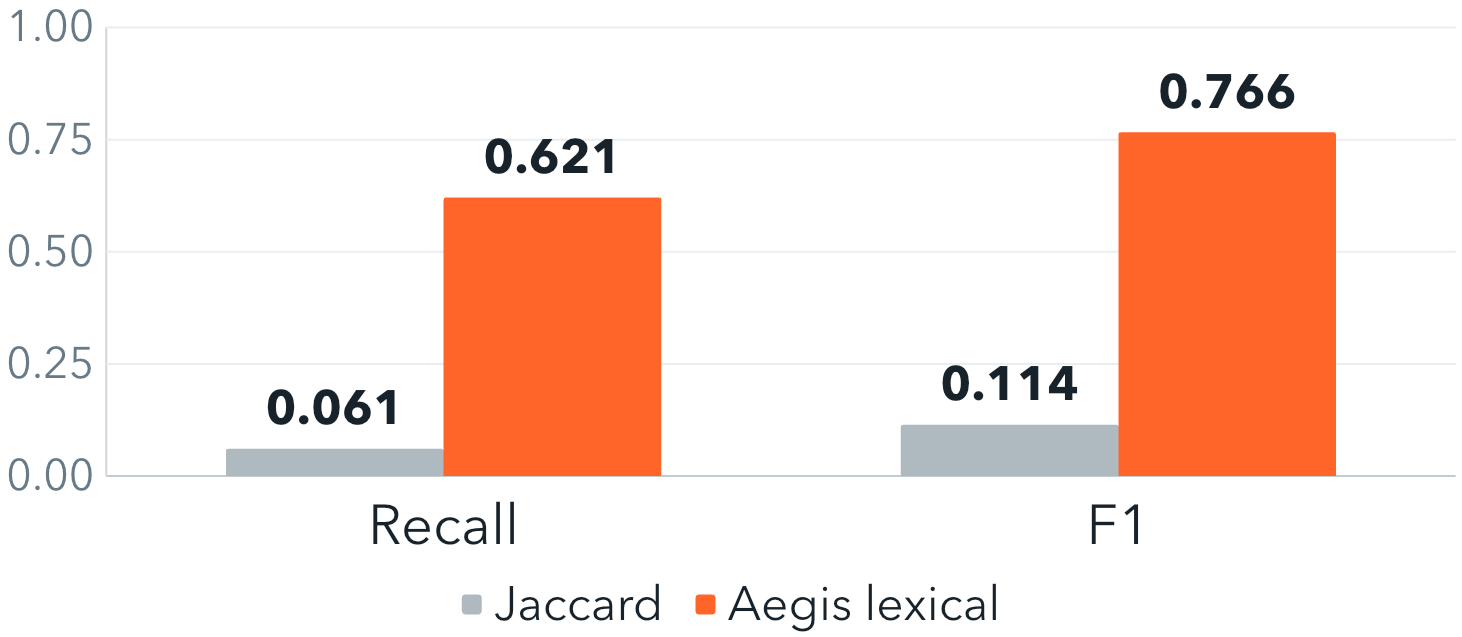
Incident families in parentheses



The fixed regression set was inspected during engineering.

Grouping results

60-report synthetic regression set



41

of 66 same-family pairs recovered

Jaccard recovered 4

1.000 precision for both methods. Zero observed false merge pairs.

Aegis missed 25 pairs. Every Aegis report still carries a review flag.

Engineering checks

196

backend tests passed

140

frontend tests passed

100% across configured coverage measures

Build, types, lint and the full Chromium exercise passed in the recorded run.

What comes next

Model

Measure real MiniLM inference
and paraphrase coverage.

Data

Evaluate on independently prepared event reports.

Use

Observe coordinators and measure correction effort.

Broader use also needs access control, operational review and device testing.

Aegis

Every accepted report stays available for coordinator review.

Live demonstration

1 **Receive and group**

2 **Inspect Exit A**

3 **Record a correction**

4 **Open evaluation**

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